

Markscheme

Mathematics: analysis and approaches

Practice question bank

9. (a)

Consider the equation $2^{2x} - 5 \times 2^x + 4 = 0$.

Find the sum of all values of x satisfying the equation.

$$\text{let } u = 2^x: u^2 - 5u + 4 = 0 \quad (M1)$$

attempt to factorise/solve the quadratic (M1)

$$(u - 1)(u - 4) = 0 \Rightarrow u = 1 \text{ or } u = 4 \quad \mathbf{A1}$$

$$2^x = 1 \Rightarrow x = 0 \quad \mathbf{A1}$$

$$2^x = 4 \Rightarrow x = 2 \quad \mathbf{A1}$$

$$\text{sum} = 2 \quad \mathbf{A1}$$

[6 marks]

(b)

Hence find the value of $2^{x_1} \times 2^{x_2}$, where x_1 and x_2 are the two solutions.

$$2^{x_1} \times 2^{x_2} = u_1 \times u_2 = 1 \times 4 \quad (M1)$$

$$= 4 \quad \mathbf{A1}$$

[2 marks]